Reproduction and Evaluation of Robust Baro–Radar–Inertial Odometry

Project Overview, Complete Theory Study Plan, and Implementation Roadmap

Research Reproduction Project

October 2026

1 Research Paper and Project Goal

Reference paper: R. Girod, M. Hauswirth, P. Pfreundschuh, M. Biasio, and R. Siegwart, "A robust baro-radar-inertial odometry m-estimator for multicopter navigation in cities and forests," 2024. Paper link.
Official implementation: https://github.com/ethz-asl/rio.

1.1 Problem addressed in the paper

A drone navigating without reliable GNSS needs estimates of its motion. IMU-only integration accumulates error; radar supplies directional Doppler motion constraints but has moving-object and multipath outliers; atmospheric pressure provides useful relative height information but is subject to disturbances. The paper proposes a robust, factor-graph-based estimator that fuses these measurements and tracks repeatable zero-velocity radar detections, especially around take-off and landing.

1.2 What the paper estimates

At each radar-frame time i, the estimated navigation state is

$$\boxed{x_i = \left(R^i_{IB},\ {}^{I}t^i_{IB},\ {}^{I}v^i_{IB},\ {}^{B}b^i_g,\ {}^{B}b^i_a\right)} \tag{3}$$

where orientation has three degrees of freedom; position, velocity, gyroscope bias, and accelerometer bias each have three components. Thus, it is a **15-dimensional local navigation state**. The algorithm can additionally estimate stationary-radar-target positions t_j for zero-velocity tracking.

1.3 Our project objective

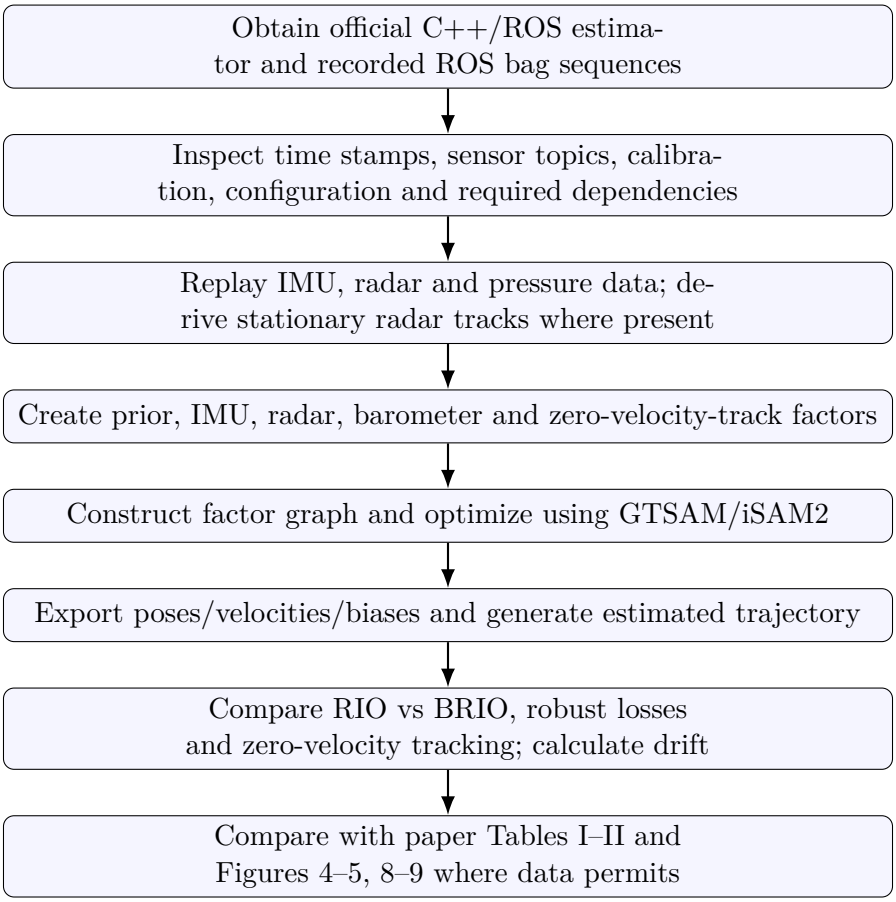
Reproduce the complete estimator's four measurement-related components—IMU preintegration, robust radar bearing-Doppler, robust barometry, and zero-velocity detection tracking—including its initialization/prior factor. Start from the authors' released code and recorded ROS bags, generate offline trajectories, then reproduce selected comparisons from the paper and test the contribution of each component.
Scope distinction: This is an *offline algorithm and evaluation reproduction*, not a reconstruction of the physical quadcopter or a reproduction of its closed-loop flight-control experiments. Exact figure-by-figure replication depends on what recorded segments, reference data, and plotting scripts are available.

2 Our Four Components and the Fifth Supporting Factor

Component	Measurements	Factor role / paper reference
IMU	Acceleration and angular rate	Preintegrated motion between states; Sec. IV-G, Eq. (15)
Radar	Detection bearing and Doppler velocity	Robust velocity constraint, Welsch loss; Sec. IV-D, Eqs. (8)–(10)
Barometer	Pressure	Relative height constraint, Fair loss; Sec. IV-E, Eqs. (11)–(12)
Zero-velocity detection tracking	Repeated radar detections at standstill	Landmark/pose consistency across frames; Sec. IV-F, Eqs. (13)–(14)
Supporting prior	Initial state and bias estimates	Anchors initial state; Sec. IV-H, Eqs. (16)–(21)

Important: Zero-velocity detection tracking is *not* a fourth sensor and is *not* simply the constraint $v = 0$. It matches stable radar detections across frames and constrains their positions relative to estimated poses.

3 End-to-End Reproduction Workflow



4 Available Resources and What Still Needs Verification

Resource	Source	Status
Paper, algorithm equations	Provided PDF, Sections III–VIII	Available
Source code	https://github.com/ethz-asl/rio	Public repository
15 sensor-recording sequences	Linked from repository; with/without camera imagery	Download links published
IMU topic	`/imu/data_raw` and initialization `/imu/data`	Documented
Radar topic	`/radar/cfar_detections`	Documented
Pressure topic	`/baro/pressure`	Optional; verify per bag
IMU–radar extrinsic calibration	Repository README	Documented
ROS Noetic, GTSAM, iSAM2	Repository setup/paper Sec. V	Required software stack
Reference end-point pose / full ground truth	Paper reports start/end alignment and benchmark metrics	Check what was actually released
Zero-velocity track occurrences	Derived from recorded radar measurements	Verify recordings and code behavior

Dataset interpretation: Each of the 15 datasets is a recording of one original experiment, not a different trained model. The estimator processes these recorded measurements; there is no machine-learning training step.

5 Complete Theory Learning Phases (Before Implementation)

Each phase specifies **what to study**, **where it appears in the paper**, and **what we must be able to explain** before moving on.

Theory Phase 1 — State Estimation Fundamentals

Read: Sec. IV-A, Eq. (3), p. 3.
Study position versus velocity; 3-D orientation; roll/pitch/yaw versus rotation matrices; gyroscope and accelerometer biases; why an IMU drifts; 15-dimensional state; difference between a state at time i and the full trajectory X_k.
Checkpoint: Explain all five state components, dimensions, and which factors constrain each.

Theory Phase 2 — Coordinate Frames and Transformations

Read: Sec. III and Fig. 2, p. 2; Eqs. (1)–(2).
Study the inertial frame I, body/IMU frame B, and radar frame R; vector coordinate conventions; rotation R_{AB}; translation ${}^{A}t_{AB}$; rigid transform T_{AB}; IMU–radar extrinsic calibration; how to project radar velocity into the radar frame.
Checkpoint: Transform one 3-D vector between radar, body and inertial coordinates and explain why calibration matters.

Theory Phase 3 — Sensors, Timestamps and Recorded Data

Read: Sec. IV-B, Eq. (5), p. 3; radar detection description in Sec. IV-D; Sec. V hardware rates.
Study IMU specific force and angular rate; radar range, direction of arrival (DoA), Doppler, SNR, noise; pressure; sparse detections versus raw radar signals; timestamps; asynchronous sampling; interpolation; what a ROS bag/topic/message contains.
Checkpoint: Describe raw inputs for one radar frame and the IMU samples between two frames.

Theory Phase 4 — IMU Kinematics and Preintegration

Read: Sec. IV-G, Eq. (15), p. 5.
Study acceleration integration, orientation integration, gravity compensation, bias correction, accumulated noise, relative changes ΔR, Δv, Δt and the five-part preintegrated IMU residual. Understand why hundreds of individual IMU measurements are combined into one factor between consecutive radar-frame states.
Checkpoint: Given IMU samples between states x_i and x_{i+1}, explain the predicted relative motion and the factor residual.

Theory Phase 5 — Radar Doppler and Ego-Velocity

Read: Sec. IV-D, Eqs. (8)–(10), pp. 3–4; Fig. 2.
Study radial velocity, sign conventions, line-of-sight unit vectors, the static-scene assumption, projection with a dot product, why 3-D radar-only velocity needs at least three independent detection directions, the radar–IMU lever arm and gyroscope-bias correction. A simplified residual is

$$\boxed{r_D^{i,m} = -\left({}^R v_{IR}^i\right)^T {}^R e_T^{i,m} - v_T^{i,m}.} \tag{8}$$

Checkpoint: Calculate a predicted Doppler for a sample motion, compare with measurement, and interpret the residual.

Theory Phase 6 — Residuals, Uncertainty and Robust Losses

Read: Sec. IV-C–D, Eq. (7), Fig. 4, pp. 3–4.
Study residual versus squared error; covariance/variance/standard deviation; normalization r/σ; Mahalanobis distance $r^T \Sigma^{-1} r$; inliers/outliers; quadratic, Huber, Fair, Cauchy and Welsch losses; why smooth robust weighting helps with moving objects and ghost radar targets; Welsch selection in the paper.
Checkpoint: Compare the influence of a small and a very large radar residual under quadratic versus Welsch losses.

Theory Phase 7 — Barometer and Robust Relative Height

Read: Sec. IV-E, Eqs. (11)–(12), pp. 4–5; Fig. 5.
Study pressure-to-height conversion; altitude offset at initialization; barometric height residual

$$\boxed{r_B^i = \left({}^I e_z\right)^T {}^I t_{IB}^i + z_p^0 - z_p^i.} \tag{11}$$

Study vertical drift, temperature/pressure bias, rotor downwash/ground effect, and the authors' Fair loss selection.
Checkpoint: Explain how a pressure sample creates an altitude constraint and what happens during a pressure disturbance.

Theory Phase 8 — Zero-Velocity Detection Tracking

Read: Sec. IV-F, Eqs. (13)–(14), p. 4.
Study why Doppler-only velocity becomes poorly constrained with only one or two returns at standstill; zero-Doppler detections; matching repeated detections based on recorded attributes; retaining/discarding target tracks; estimated stationary target location t_j; coordinate transformation and landmark residual.
Checkpoint: Draw two drone states observing the same stationary radar target; explain why this adds geometric constraints. Distinguish this approach from a direct $v = 0$ prior.

Theory Phase 9 — Initialization and Prior Factor

Read: Sec. IV-H, Eqs. (16)–(21), p. 5.
Study initial orientation from filtered IMU, early gyroscope bias averaging, initialization of position/velocity/accelerometer bias, prior uncertainty, and the initial-state anchor.
Checkpoint: Explain what initializes the optimizer and why poor yaw bias initialization can produce drift.

Theory Phase 10 — Factor Graphs and the Complete Objective

Read: Sec. IV-C, Fig. 3, Eqs. (6)–(7), p. 3.
Study variable nodes (states x_i and tracked target positions t_j), factor nodes, connectivity, MAP estimation, residual construction, covariances, robust terms, and how constraints accumulate. Learn that factors are assembled *before/during* optimization; their values are not sent into a separate graph afterward. The objective consists of

$$\boxed{J = J_{\text{prior}} + J_{\text{IMU}} + J_{\text{track}} + J_{\text{radar,robust}} + J_{\text{baro,robust}}.} \qquad (7, \text{schematic})$$

Checkpoint: Draw a 3-state factor graph with each applicable factor, specify its variables, and follow its measurements into the optimizer.

Theory Phase 11 — Numerical Optimization, Sliding Window and iSAM2

Read: Sec. IV-C and Sec. V, p. 5.
Study nonlinear least squares, linearization, iterative state updates, iSAM2 incremental optimization, 10-second window in the paper, uncertainty weighting, and how the navigation thread predicts states at IMU rate. Clarify the difference between optimizer output and controller feedback.
Checkpoint: Explain the full sequence: sensor measurement → residual → robust cost → factor graph → optimizer → updated state.

Theory Phase 12 — Evaluation, Ablations and Limitations

Read: Secs. V–VII, Tables I–II, Figs. 4–9, pp. 5–8.
Study final position drift, drift/distance, XY drift, height drift, yaw drift, relative pose error (RPE); handheld versus flown sequences; RIO (without barometer) versus BRIO (with barometer); streetcar/multipath/pressure examples; the external ColoRadar benchmark (without barometer data); limitations such as yaw observability, bearing bias and vibration.
Checkpoint: Interpret Table I and distinguish claims supported by paper results from claims requiring our own experiments.

6 Implementation Phases (After Theory)

Phase	Activity	Deliverable / completion check
I1	Environment and source audit	Install a compatible ROS Noetic environment, clone the official packages, record commit hashes, build successfully, understand launch/config files.
I2	Data audit	Download one image-free ROS bag, inspect radar/IMU/barometer topics, frequencies, timestamps and calibration. Determine how estimator outputs and reference poses will be obtained.

Phase	Activity	Deliverable / completion check
I3	Run official RIO baseline	Replay Dataset 01 with barometer disabled, export estimated trajectory and confirm the process is stable.
I4	Run complete BRIO	Enable barometer on the same dataset; verify all intended factors including zero-velocity tracking are active where observations permit.
I5	Test factor contributions	Where settings/code permit, test radar+IMU, radar+IMU+barometer, radar+IMU+tracking, and all components together. Keep unrelated settings constant.
I6	Compare robust losses	Run repeated configurations with different radar robust losses; compare estimates during moving-object segments and inspect residuals.
I7	Reproduce evaluation	Export trajectories; write scripts for applicable drift, yaw, and RPE metrics; document missing ground-truth limitations.
I8	Generalization	Repeat selected urban, forest and flying sequences; compare to original paper Tables I–II and figures where feasible.
I9	Final analysis	Create tables, plots, setup log, reproduction limitations, and final report.

7 Proposed Experiments and Outputs

Experiment	What changes?	What to compare?
RIO vs BRIO	Barometer factor disabled/enabled	Estimated height and final drift; paper Table I and Fig. 8
Radar robust-loss test	Quadratic, Huber, Fair, Cauchy, Welsch	Velocity stability and trajectory error; paper Fig. 4 (exact segment if available)
Zero-velocity tracking test	Tracking enabled/disabled if supported	Stationary-period position/velocity drift
Multi-environment test	Dataset sequence	Urban/forest/flown trajectory stability and drift
ColoRadar extension (optional)	Alternative public dataset (no barometer)	Paper Section VI and Table II, radar-inertial only

Target example from Table I: Dataset 01 (Urban Night, handheld) reports RIO final position drift 3.54 m (1.97% of distance) and BRIO 1.42 m (0.77%). These are *reference results*, not results we have already reproduced.

8 Final Deliverables

1. **Reproducible setup:** documented ROS/C++/GTSAM environment, exact repository revision and configuration settings.
2. **Working offline estimator:** demonstration of IMU, radar, barometer and zero-velocity tracking in the available recorded data.

3. **Trajectory outputs:** exported estimated 3-D trajectories and orientation/velocity plots.
4. **Comparison experiments:** RIO/BRIO, radar loss comparisons and zero-velocity tracking analysis when supported by recordings and settings.
5. **Quantitative evaluation:** drift and applicable pose metrics compared against published tables.
6. **Final report:** methods, plots, reproduced results, differences, limitations and conclusions.

9 Precise Project Description for the Professor

Our project will reproduce the robust baro-radar-inertial odometry estimator proposed by Girod et al. We will use the authors' published C++/ROS source code and recorded radar, IMU, and pressure datasets. The algorithm employs IMU preintegration, robust bearing-Doppler radar factors, robust barometer factors, and zero-velocity radar detection tracking, plus initialization priors. We will first study the theory and equations for each factor, then build and run the original software on the released ROS bags. We will export trajectory estimates, compare configurations such as RIO versus BRIO and different radar loss functions, and calculate applicable drift metrics to compare with the original publication. We are reproducing the estimation algorithm and offline evaluation, not rebuilding the drone or repeating physical flight trials.

10 How We Will Know We Are Ready to Implement

Before writing or changing code, we should be able to answer all of these:

1. What is the 15-dimensional state and what does each component represent?
2. How are inertial, IMU/body, and radar frames connected?
3. What do radar, IMU, and pressure messages contain and at what times?
4. How does IMU preintegration relate two consecutive states?
5. How does a measured Doppler value create a radar factor residual?
6. How do standard deviations, covariance and robust loss alter the cost?
7. How is pressure converted to a height residual?
8. How does tracking a fixed radar detection constrain standstill states?
9. Why do we need a prior factor and bias initialization?
10. How are all the factors combined and optimized in Eq. (7)?
11. How do we compute and interpret paper-style trajectory drift?
12. Which measurements and reference values are actually present in our downloaded ROS bags?

11 References and Links

1. Girod et al., 2024: https://arxiv.org/abs/2408.05764.
2. Official estimator and dataset links: https://github.com/ethz-asl/rio.
3. Paper Sections IV–VI, especially Eqs. (3), (7)–(15), Figs. 3–5 and Tables I–II.